



Report Title:

TryHackMe, Linux

Fundamentals Part 1

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Prepared for: Readers and Beginners

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Task 1: Introduction

Welcome to the world of Linux fundamentals! In this TryHackMe room, I'll guide you through essential commands and operators of Linux.

Answer the given question below

Let's get started!

Ans. Just click on Submit! (*no answer needed*)

A screenshot of a dark-themed interface. At the top, it says "Answer the questions below" in green. Below that, it says "Let's get started!". There is a grey input field containing the text "No answer needed". To the right of the input field is a bright green button with a white checkmark and the text "Correct Answer".

Answer the questions below

Let's get started!

No answer needed

✓ Correct Answer

Task 2: A Bit of Background on Linux

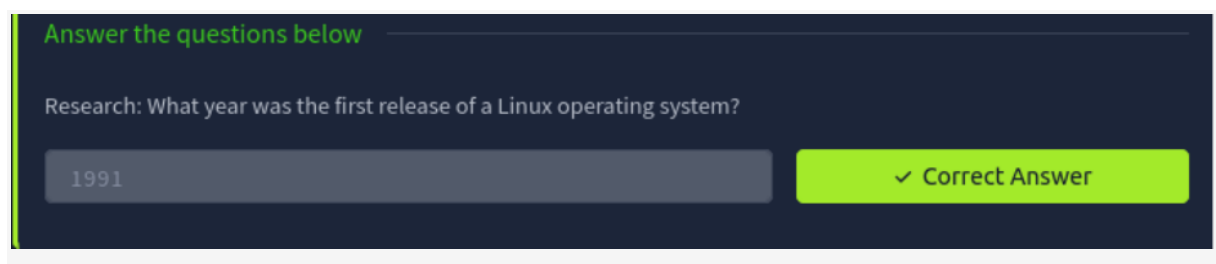
Explore the roots of Linux, a lightweight, open-source, secure, stable, and flexible OS created by Linus Torvalds

in 1991. Discover its popularity and various distributions, including Kali Linux, Ubuntu, Fedora, and Arch.

Answer the given question below

Research: What year was the first release of a Linux operating system?

Ans. 1991



Answer the questions below

Research: What year was the first release of a Linux operating system?

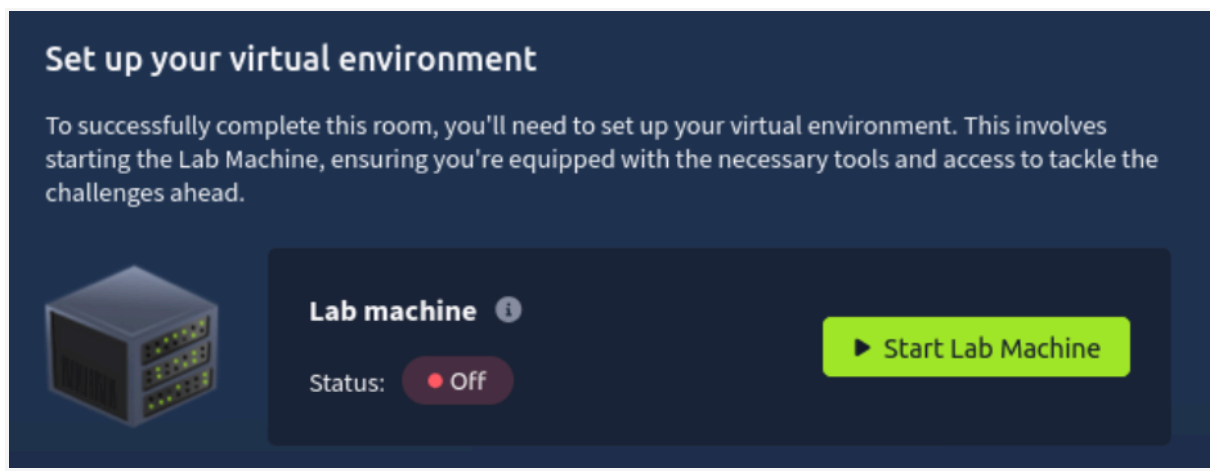
1991

✓ Correct Answer

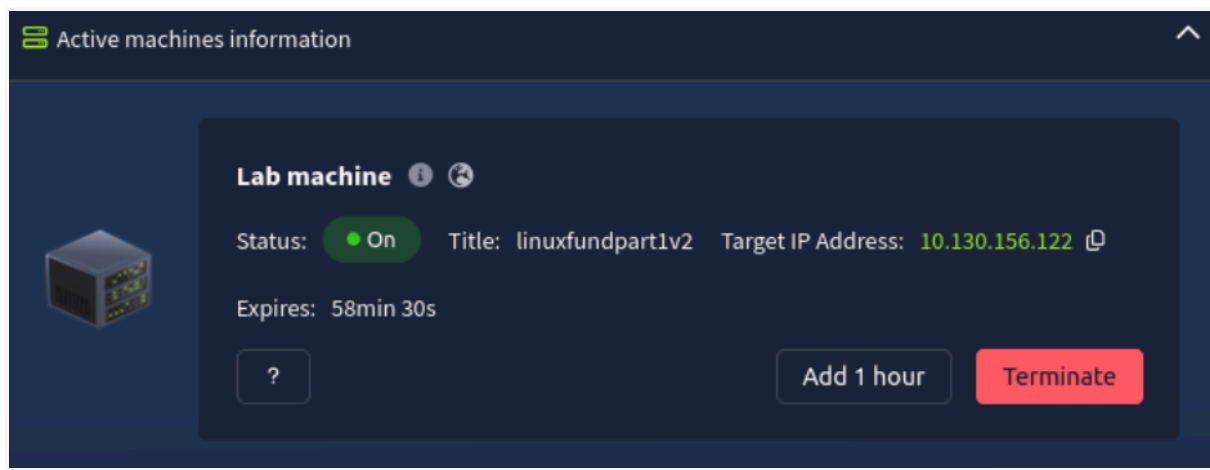
Task 3: Interacting With Your First Linux

Machine (In-Browser)

Embark on your Linux journey by deploying your first machine on TryHackMe. Receive an IP address and a termination time, and explore the Ubuntu distribution. No answers needed here, just enjoy the process!



And Boom! Your machine is Started. You will be given an IP address and the termination of your machine.



The Linux distribution on TryhackMe is Ubuntu.

Answer the given question below

I've deployed my first Linux machine!

Answer the questions below

I've deployed my first Linux machine!

No answer needed

✓ Correct Answer

Task 4: Running Your First Few Commands

The two basic commands discussed in Task 4 are **echo** and **whoami**.

echo: In simple words, echo works just like how print command works in different languages. For example, if you enter **\$ echo "hello world"** it will print hello world.

```
tryhackme@linux1:~$ echo "hello world"
hello world
tryhackme@linux1:~$
```

whoami: This command will give you the name of the user with which you are currently

logged in. We will see an example of whoami in task 4 questions.

Answer the given question below

If we wanted to output the text **“TryHackMe”**, what would our command be?

```
tryhackme@linux1:~$ echo "TryHackMe"  
TryHackMe  
tryhackme@linux1:~$ █
```

Ans. Echo TryHackMe

What is the username of who you're logged in as on your deployed Linux machine?

```
tryhackme@linux1:~$ whoami  
tryhackme  
tryhackme@linux1:~$ █
```

Ans. tryhackme

Answer the questions below

If we wanted to output the text "TryHackMe", what would our command be?

✓ Correct Answer 

What is the username of who you're logged in as on your deployed Linux machine?

✓ Correct Answer 

Task 5: Interacting With the Filesystem!

The commands discussed in task 5 are **ls** , **cd** , **cat** , **pwd**

ls: The most basic command in linux is ls. It will list all the directories and files of your home directory, or any other directory you are in (Think directory as a folder). For example: Perform ls command on your machine and see the output.


```
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$
```

It lists for you all the files and folders which are stored in your machine.

cd: The next command is `cd` which changes the directory. For example, you want to move in some directory say `folder1` then you will perform a command **`cd folder1`** and now you are in `folder1`

```
tryhackme@linux1:~$ cd folder1
tryhackme@linux1:~/folder1$
```

Notice the change of address when we moved in `folder1`

cat: Suppose you want to see the content of any file. In that case, you will use the command `cat`. For example: I want to see the

content of file1, in that case I will perform **cat file1**.

```
tryhackme@linux1:~$ cat file1  
hello world  
tryhackme@linux1:~$ █
```

pwd: pwd stands for print working directory. It will print you the address of the directory you are in.

```
tryhackme@linux1:~$ pwd  
/home/tryhackme
```

Answer the given question below

On the Linux machine that you deploy, how many folders are there?

```
tryhackme@linux1:~$ ls  
access.log  folder1  folder2  folder3  folder4  
tryhackme@linux1:~$ █
```

Ans. 4

Which directory contains a file?

In this, we need to move in each directory and check if there is a file or not.

```
tryhackme@linux1:~$ cd folder1
tryhackme@linux1:~/folder1$ ls -la
total 8
drwxr-xr-x 2 tryhackme tryhackme 4096 May 10  2021 .
drwxr-xr-x 8 tryhackme tryhackme 4096 Jul 19 12:07 ..
tryhackme@linux1:~/folder1$ cd ..
tryhackme@linux1:~$ cd folder2
tryhackme@linux1:~/folder2$ ls -la
total 8
drwxr-xr-x 2 tryhackme tryhackme 4096 May 10  2021 .
drwxr-xr-x 8 tryhackme tryhackme 4096 Jul 19 12:07 ..
tryhackme@linux1:~/folder2$ cd folder3
-bash: cd: folder3: No such file or directory
tryhackme@linux1:~/folder2$ ls -la
total 8
drwxr-xr-x 2 tryhackme tryhackme 4096 May 10  2021 .
drwxr-xr-x 8 tryhackme tryhackme 4096 Jul 19 12:07 ..
tryhackme@linux1:~/folder2$ cd ..
tryhackme@linux1:~$ cd folder4
tryhackme@linux1:~/folder4$ ls -la
total 12
drwxr-xr-x 2 tryhackme tryhackme 4096 May 10  2021 .
drwxr-xr-x 8 tryhackme tryhackme 4096 Jul 19 12:07 ..
-rw-r--r-- 1 tryhackme tryhackme  13 May 10  2021 note.txt
tryhackme@linux1:~/folder4$ █
```

Ans. folder4

What are the contents of this file?

```
tryhackme@linux1:~/folder4$ cat note.txt
Hello World!
tryhackme@linux1:~/folder4$ █
```

Ans. Hello World!

Use the cd command to navigate to this file and find out the new current working directory. What is the path?

```
tryhackme@linux1:~/folder4$ pwd  
/home/tryhackme/folder4  
tryhackme@linux1:~/folder4$
```

Ans. /home/tryhackme/folder4

Answer the questions below

On the Linux machine that you deploy, how many folders are there?

4

✓ Correct Answer

Which directory contains a file?

folder4

✓ Correct Answer 0

What is the contents of this file?

Hello World!

✓ Correct Answer

Use the cd command to navigate to this file and find out the new current working directory. What is the path?

/home/tryhackme/folder4

✓ Correct Answer

Task 6: Searching for files

In this task, we will discuss about two commands **find** and **grep**

find: Suppose you want to search for a file among thousands of directories, how will you find it? In this situation, find comes into action. Find — allows us to search for files or directories that match specific search criteria. Two common options are -type and -name. For -type, we can use d for directories or f for files, for example:

```
find -type d
```

```
find -type f
```

If you want to search for note.txt in your machine you will write, **find -type f -name note.txt**

```
tryhackme@linux1:~$ find -type f -name note.txt  
./folder4/note.txt  
tryhackme@linux1:~$
```

And suppose you do not know the full name of the file, but you know the extension, in that case you can use **find -name *.txt**

```
tryhackme@linux1:~$ find -name *.txt  
./folder4/note.txt  
tryhackme@linux1:~$
```

grep: grep allows us to search the content of files. We will see the use of the grep command in the questions.

wc helps us to know the number of lines in a file.

```
tryhackme@linux1:~$ wc -l access.log  
302 access.log  
tryhackme@linux1:~$
```

Answer the question below

Use grep on “access.log” to find the flag that has a prefix of “THM”. What is the flag?

```
tryhackme@linux1:~$ grep "THM" access.log
13.127.130.212 - - [04/May/2021:08:35:26 +0000] "GET THM{ACCESS}
  lang=en HTTP/1.1" 404 360 "-" "Mozilla/5.0 (Windows NT 10.0; Wi
n64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/77.0.386
5.120 Safari/537.36"
tryhackme@linux1:~$
```

Ans. THM{ACCESS}

And I still haven't found what I'm looking for!

Ans . No answer needed

Answer the questions below

Use grep on "access.log" to find the flag that has a prefix of "THM". What is the flag?
Note: The "access.log" file is located in the "/home/tryhackme/" directory.

THM{ACCESS}

✓ Correct Answer

And I still haven't found what I'm looking for!

No answer needed

✓ Correct Answer

Task 7: An Introduction to Shell Operators

In this task, we will discuss shell operators like **&**, **&&**, **>**, **>>**

&: Suppose you are copying a large file on linux and you also want to do other things while your file is being copied. So, you can use **&** operator to keep using the terminal while the file is being copied in background.

&&: You can use this operator where you want to combine two or more commands. Note that command 2 will only run if command 1 is successful.

For Example: I want to move in folder4 and print the content of note.txt.


```
tryhackme@linux1:~$ cd folder4 && cat note.txt
Hello World!
tryhackme@linux1:~/folder4$
```

It will first move in folder4 and then print the content of the file.

>: This operator is known as an output redirector. Suppose you want to store the output of some command in some file, so you can use this operator.

```
tryhackme@linux1:~$ cd folder4
tryhackme@linux1:~/folder4$ ls
note.txt
tryhackme@linux1:~/folder4$ echo Linux Fundamentals Part 1 > Linux.txt
tryhackme@linux1:~/folder4$ ls
Linux.txt  note.txt
tryhackme@linux1:~/folder4$ cat Linux.txt
Linux Fundamentals Part 1
tryhackme@linux1:~/folder4$ 4
```

As you can see, it automatically creates a file with the name you provided.

>> If you want to add another output in the same file, you can use this operator.

```
tryhackme@linux1:~/folder4$ echo Hello >> Linux.txt
tryhackme@linux1:~/folder4$ ls
Linux.txt  note.txt
tryhackme@linux1:~/folder4$ cat Linux.txt
Linux Fundamentals Part 1
Hello
tryhackme@linux1:~/folder4$ █
```

As you can see the output is added at the end of the file .

Answer the questions below

If we wanted to run a command in the background, what operator would we want to use?

Ans: &

If I wanted to replace the contents of a file named “passwords” with the word “password123”, what would my command be?

```
tryhackme@linux1:~/folder4$ echo password123 >> passwords
tryhackme@linux1:~/folder4$ ls
Linux.txt  note.txt  passwords
tryhackme@linux1:~/folder4$ cat passwords
password123
tryhackme@linux1:~/folder4$ █
```

Ans: `echo password123 > passwords`

Now if I wanted to add “tryhackme” to this file named “passwords” but also keep “passwords123”, what would my command be?

```
tryhackme@linux1:~/folder4$ echo tryhackme >> passwords
tryhackme@linux1:~/folder4$ ls
Linux.txt  note.txt  passwords
tryhackme@linux1:~/folder4$ cat passwords
password123
tryhackme
tryhackme@linux1:~/folder4$ █
```

Ans: `echo tryhackme >> passwords`

Now use the deployed Linux machine to put these into practice.

Ans. No answer needed

Answer the questions below

If we wanted to run a command in the background, what operator would we want to use?

&

✓ Correct Answer

If I wanted to replace the contents of a file named "passwords" with the word "password123", what would my command be?

echo password123 > passwords

✓ Correct Answer

Now if I wanted to add "tryhackme" to this file named "passwords" but also keep "passwords123", what would my command be?

echo tryhackme >> passwords

✓ Correct Answer

Now use the deployed Linux machine to put these into practice

No answer needed

✓ Correct Answer

Task 8: Conclusions & Summaries

We have covered some commands of Linux with examples. They may seem basic, but they are essential for mastery of Linux, and we will see them again.

Answer the questions below

I'll have a play-around!

Ans. No answer needed



Task 9: Linux Fundamentals Part 2

If you want to learn more commands of linux, please join Linux Fundamentals Part 2 with following link: <https://tryhackme.com/room/linuxfundamentalspart2>

Answer the questions below

Terminate the machine deployed in this room from task 3.

Ans. No answer needed

Join Linux Fundamentals Part 2!

Ans. No answer needed

Task 10: Conclusion

In conclusion, this report has presented the main findings and outcomes of the project. The completed work demonstrates that the objectives were successfully achieved through careful planning, implementation, and evaluation. The results indicate that the proposed approach is effective and meets the required expectations. Furthermore, the experience gained throughout this task provides valuable insights for future improvements and similar projects. Overall, the project highlights the importance of continuous development, proper analysis, and the application of best practices to achieve reliable and high-quality results.